it practical for real-world deployment. By capturing the underlying structure of user preferences, LoRe generalizes effectively to unseen users with minimal data. Unlike latent-code-based methods like VPL or PAL that require separate modules for inferring user representations, LoRe directly learns a compact basis, reducing the number of learnable parameters and improving both efficiency and generalization.

3. **Scalability to Large and Diverse User Populations** Unlike approaches that either assume homogeneous reward models (BT) or struggle with scalability (PAL, VPL), LoRe maintains strong performance on large-scale personalization tasks. The low-rank decomposition reduces computational overhead while preserving expressiveness (cf. Sec. 5).

4. **Integration with Multi-Objective Alignment for Response Generation** LoRe naturally extends to personalized response generation by leveraging its basis reward functions. Unlike PAL and VPL, which require additional policy networks to generate responses, our method integrates seamlessly with steerable multi-objective alignment frameworks.

5. **Bridging the Gap Between Explicit Categorization and Per-User Models** Many personalization methods either cluster users into predefined categories (demographics, personality types) or train separate models for each user. LoRe avoids these extremes by learning a flexible, data-efficient representation of user preferences that adapts without requiring extensive individual data.

By combining structured reward decomposition, scalable adaptation, and efficient integration with response generation, LoRe offers a principled and practical approach to personalized RLHF, addressing key limitations of prior methods while enabling new capabilities.

# 5 Experiments

**Evaluation Metrics:** We evaluate the reward model's accuracy on unseen response pairs for both seen and unseen users:

$$\frac{1}{|\tilde{\mathcal{D}}_i|} \sum_{(x,y_c,y_r)\in\tilde{\mathcal{D}}_i} \mathbb{I}[\mathbf{w}_i^\top(\mathbf{R}_\phi(x,y_c) - \mathbf{R}_\phi(x,y_r)) > 0], \tag{15}$$

where $\mathbb{I}[\cdot]$ is the indicator function, equal to 1 if the condition holds and 0 otherwise.

We define four dataset splits: $\mathcal{D}_{\text{train}}$ contains labeled preference data from seen users $\mathcal{U}_{\text{seen}}$ used to train the reward basis, $\mathcal{D}_{\text{test}}^{\text{seen}}$ evaluates generalization to new response pairs for seen users, $\mathcal{D}_{\text{fewshot}}$ provides a small set of labeled data for unseen users $\mathcal{U}_{\text{unseen}}$, and $\mathcal{D}_{\text{test}}^{\text{unseen}}$ assesses generalization to new users. We evaluate:

1. **Seen Accuracy:** Generalization to new response pairs for seen users using $\mathcal{D}_{\text{test}}^{\text{seen}}$.

2. **Unseen Accuracy:** Generalization to new response pairs for unseen users on $\mathcal{D}_{\text{test}}^{\text{unseen}}$, where user preferences are learned from few-shot examples in $\mathcal{D}_{\text{fewshot}}$.

3. **Few-shot Ability:** Accuracy on $\mathcal{D}_{\text{test}}^{\text{unseen}}$ upon varying the number of examples in $\mathcal{D}_{\text{fewshot}}$.

**Baselines:** We use a pre-trained reward model (Liu et al., 2024) to generate fixed embeddings $\mathbf{e}(x,y) \in \mathbb{R}^D$, where $D = 4096$. We compare against 1) **Reference Model** (Liu et al., 2024), 2) **BT** (monolithic reward model, applying a learnable linear mapping from fixed embeddings $\mathbf{e}(x,y)$ to a scalar reward), 3) **VPL** (Poddar et al., 2024), 4) **PAL** (Chen et al., 2024a) (both using their respective architectures over the same fixed embeddings $\mathbf{e}(x,y)$), and 5) **LoRe** (applying a learnable linear transformation on $\mathbf{e}(x,y)$ to map embeddings to $\mathbb{R}^B$ (corresponding to Example 1 in Sec. 3.2). A detailed description of all baselines is presented in Appendix B.1.

**Semi-synthetic Preference Dataset:** The PersonalLLM dataset (Zollo et al., 2024) contains 10,402 prompts, each with responses from eight top LLMs (e.g., GPT-4o, Claude 3 Opus, Mixtral8x22B). Each response is scored by 10 reward models from Reward Bench, built on popular base models such as Llama3, Mistral, and Gemma, with diverse preferences. Given a prompt $x$ and response $y$, the reward vector is $\mathbf{R}(x,y) \in \mathbb{R}^{10}$. The dataset is split into 9,402 training and 1,000 test prompts.

| Setting | PersonalLLM | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Very Diverse ($\alpha = 0.001$) | | | Moderately Diverse ($\alpha = 0.01$) | | | Near Uniform ($\alpha = 0.1$) | | |
| **Method** | **Seen** | **Unseen** | **Overall** | **Seen** | **Unseen** | **Overall** | **Seen** | **Unseen** | **Overall** |
| ref | 78.4 ± 1.2 | 76.1 ± 1.5 | 77.3 ± 1.3 | 78.1 ± 1.4 | 77.8 ± 1.6 | 77.9 ± 1.5 | 82.7 ± 1.0 | 83.7 ± 1.1 | 83.2 ± 1.0 |
| BT | 86.3 ± 1.0 | 86.4 ± 1.3 | 86.4 ± 1.1 | 87.8 ± 1.2 | 87.2 ± 1.4 | 87.5 ± 1.3 | 93.2 ± 0.9 | 93.1 ± 1.0 | 93.2 ± 0.9 |
| VPL | 86.4 ± 1.3 | 86.5 ± 1.2 | 86.5 ± 1.3 | 93.9 ± 1.1 | 84.1 ± 1.5 | 89.0 ± 1.3 | 92.0 ± 1.0 | 92.8 ± 0.8 | 92.4 ± 0.9 |
| PAL | 85.0 ± 1.3 | 86.5 ± 1.2 | 85.7 ± 1.3 | 86.1 ± 1.1 | 87.1 ± 1.5 | 86.6 ± 1.3 | 91.7 ± 1.0 | 91.8 ± 0.8 | 91.7 ± 0.9 |
| LoRe | **94.3** ± 0.9 | **93.3** ± 1.0 | **93.8** ± 0.9 | **94.6** ± 0.8 | **93.6** ± 1.1 | **94.1** ± 0.9 | **96.0** ± 0.7 | **96.1** ± 0.8 | **96.0** ± 0.7 |

**Table 1** Using **PersonalLLM** we generate 1000 seen users and 1000 unseen users. In particular, we use 45 examples per seen user and 9 few-shot examples per unseen user.

Synthetic users are generated by sampling a preference vector $\mathbf{w} \sim \text{Dirichlet}(\alpha)$ and computing response scores as $\mathbf{w}^\top \mathbf{R}(x, y)$. We vary $\alpha$ in the range $\{0.1, 0.01, 0.001\}$, where a larger $\alpha$ results in more uniform preferences and a smaller $\alpha$ leads to more discrete user types. We then categorize users as follows: 1. **Very Diverse** ($\alpha = 0.001$): Aligning closely with one of the 10 reward models. 2. **Moderately diverse** ($\alpha = 0.01$): A balance between specific and broad preferences. 3. **Near Uniform** ($\alpha = 0.1$): The most uniform preferences.

For each user, we store the highest/lowest scored responses and simulate 1000 seen and 1000 unseen users. Each seen user gets 45 prompts from the training set $\mathcal{D}_{\text{train}}$, while each unseen user gets 9 prompts to form $\mathcal{D}_{\text{fewshot}}$. All 1000 test prompts form $\mathcal{D}_{\text{test}}^{\text{seen}}, \mathcal{D}_{\text{test}}^{\text{unseen}}$ to test the performance of the learnt models.

**Summarization Task on Real Users:** We use the TLDR dataset, where each comparison consists of a Reddit post, two summaries, and the worker ID who annotated it (Stiennon et al., 2020). After filtering out workers with fewer than 50 annotations, we retain 40 workers. They are split into two equal groups of 20, corresponding to $\mathcal{U}_{\text{seen}}$ and $\mathcal{U}_{\text{unseen}}$.

| Setting | Reddit TLDR | | | | | |
|---|---|---|---|---|---|---|
| | 100 examples per seen user | | | 150 examples per seen user | | |
| **Method** | **Seen** | **Unseen** | **Overall** | **Seen** | **Unseen** | **Overall** |
| ref | 56.3 ± 1.3 | 57.3 ± 1.4 | 56.8 ± 1.2 | 56.3 ± 1.2 | 57.3 ± 1.5 | 56.8 ± 1.3 |
| BT | 60.0 ± 1.4 | 60.0 ± 1.2 | 60.0 ± 1.3 | 63.2 ± 1.1 | 64.3± 1.3 | 63.7 ± 1.2 |
| VPL | 63.6 ± 1.3 | 62.1 ± 1.4 | 62.9 ± 1.2 | 63.4 ± 1.3 | 62.7 ± 1.2 | 63.1 ± 1.3 |
| PAL | 64.1 ± 1.1 | 64.9 ± 1.5 | 64.5 ± 1.2 | 64.4 ± 1.3 | 63.8 ± 1.2 | 64.1 ± 1.3 |
| LoRe | **65.0** ± 1.1 | **66.2** ± 1.2 | **65.6** ± 1.1 | **66.2** ± 1.0 | **66.7** ± 1.1 | **65.5** ± 1.0 |

**Table 2** We split **Reddit TLDR** into 20 seen and 20 unseen users, with 50 few-shot examples per unseen user. We vary the number of examples per seen user to learn the reward basis.

Each worker has an average of 4,000 labeled pairs, which are evenly divided into a training set and a test set. To construct $\mathcal{D}_{\text{train}}$, we randomly sample $\{100, 150\}$ pairs from each seen user's training data. For unseen users, we randomly select 50 examples from their training data to form $\mathcal{D}_{\text{fewshot}}$. Evaluation is conducted on the full set of labeled test samples, corresponding to $\mathcal{D}_{\text{test}}^{\text{seen}}$ and $\mathcal{D}_{\text{test}}^{\text{unseen}}$.

**Real-World Preference Dataset with a Large and Diverse User Base** The PRISM dataset (Kirk et al., 2024b) is a comprehensive resource for LLM feedback analysis, featuring 1,500 participants from 75 countries. It provides fine-grained feedback on both contextual and stated preferences, collected from 8,011 live conversations across 21 different LLMs.

After filtering out users with fewer than six dialogues, we randomly split the remaining participants into two equal groups, resulting in $|\mathcal{U}_{\text{seen}}| = |\mathcal{U}_{\text{unseen}}| = 643$ users, with each user averaging seven dialogues. For each user, half of their interactions are used for training $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{fewshot}})$, which informs reward model learning and the preferences of unseen users. The remaining interactions are reserved for evaluation $(\mathcal{D}_{\text{test}}^{\text{seen}}, \mathcal{D}_{\text{test}}^{\text{unseen}})$.

We repeat each experiment 20 times with different random splits, reporting the mean and standard deviation in Tables 1, 2, and 3. To further analyze the scalability of the few-shot adaptation phase, we vary the number of few-shot samples and plot the average performance across 20 runs in Figure 2.

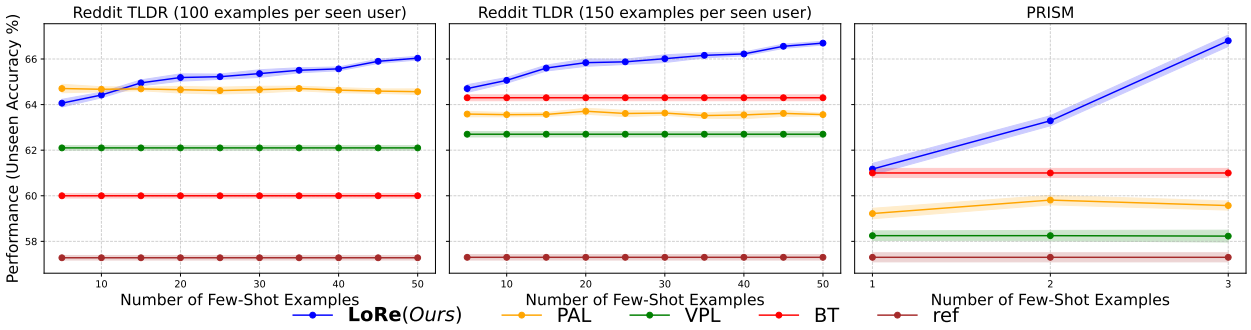


**Figure 2** We vary the number of few-shot samples and repeat each experiment 20 times, randomly subsampling different examples in each run. The plot reports the average performance (unseen accuracy) along with standard deviations. Notably, VPL, which infers the latent code from few-shot examples without the ability to relearn it, shows limited improvement as the number of examples increases. While PAL exhibits some gains, our algorithm's performance improves significantly faster in comparison.

| | PRISM | | |
|---|---|---|---|
| **Method** | **Seen** | **Unseen** | **Overall** |
| ref | 58.8 ± 1.1 | 57.3 ± 1.2 | 58.0 ± 1.0 |
| BT | 64.0 ± 1.0 | 61.0 ± 1.2 | 62.5 ± 1.1 |
| VPL | 64.6 ± 1.0 | 58.2 ± 1.2 | 61.4 ± 1.1 |
| PAL | 70.8 ± 1.0 | 59.0 ± 1.2 | 64.92 ± 1.1 |
| LoRe | **71.0** ± 0.9 | **71.0** ± 1.0 | **71.0** ± 0.8 |

**Table 3** We split **PRISM** into 643 seen and 643 unseen users. On average there are only 3.84 examples per seen and 3.87 examples per unseen user, making generalization to unseen users challenging for baselines.

**Analysis** In the PersonalLLM dataset, as diversity decreases, all methods improve, but LoRe consistently achieves the best performance across all settings. VPL and PAL, which are personalization baselines, struggle to remain competitive when the number of users is large, as seen in both PersonalLLM and PRISM. Their performance is often close to BT, which does not personalize at all, indicating a lack of scalability. However, when the number of users is smaller, as in the Reddit TLDR dataset, both VPL and PAL perform well, further reinforcing their limitations in scaling to larger personalization tasks. Additionally, VPL exhibits signs of overfitting, performing well on seen users but significantly worse on unseen users, highlighting its inability to generalize effectively. In contrast, LoRe consistently outperforms all other methods, demonstrating strong scalability, generalization, and adaptability across varying levels of personalization diversity.

We also analyze the dependence of few-shot examples on unseen accuracy in Figure 2. The BT and ref models are not capable of personalization to new users and hence demonstrate the same performance throughout. We observe that VPL and PAL do not change their performance much either as number of few shot examples increase. We found that the high-dimensional latent codes do not change significantly for both VPL and PAL as the number of few-shot examples increased. For LoRe, the unseen accuracy steadily increases across all datasets, as the number of examples increases.

## 6 Conclusion and Future Work

We introduced LoRe, a novel framework for personalizing LLMs via Low-Rank Reward Modeling. Our approach improves RLHF personalization by leveraging a structured decomposition of reward functions, enabling efficient adaptation to diverse user preferences with minimal data. Extensive evaluations demonstrated LoRe's superior generalization to seen and unseen users while maintaining scalability and efficiency. Compared to baseline methods, LoRe consistently achieved better unseen user adaptation and preference prediction accuracy. It remains effective even as the number of users increases, breaking a key limitation in prior work. Future directions include extending LoRe to online RLHF with explorative data collection.